

Code-Layout, Readability and Reusability

Date	14 July, 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This report assesses our Credit Card Approval Predictor's codebase for structural integrity, clarity, ease of maintenance, and component reusability. Built with a hybrid ExpressJS backend and Python machine learning stack, the application adheres to strict coding conventions, modular design principles, and comprehensive documentation to ensure it remains scalable and accessible for future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Uniform Code Formatting	Uniform indentation and spacing maintained throughout the Python code.	Yes	Enhances code navigation
2	Organized Directory Architecture	Assets are logically separated into distinct directori ML models, and raw datasets.	Yes	Adheres to standard Node/ Python project layouts.
3	Descriptive Naming Conventions	Variables such as applicant_income, credit_history, employment_tenure, and approval_status explicitly	Yes	Easy to understand
4	Modular Function Design	Functions like predict(), load_model(), and preprocess_data() are descriptively named.	Yes	Enhances maintainability.
5	Code comments	Important sections contain comments explaining preprocessing, model loading, and prediction logic.	Yes	Helps future developers.
6	Modular Design	Data preprocessing, model training, evaluation, and web application are separated into different modules.	Yes	Highly reusable.
7	No Redundant Code	Duplicate code has been minimized by using reusable functions.	Yes	Cleaner implementation.

8	Error Handling	Invalid user inputs and prediction errors are handled using exception handling.	Yes	Prevents application crashes.
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Used	Reusability Level (High / Medium / Low)
1	Flask Web Application	Flask, HTML, CSS	User Interface	High
2	Feature Encoding Module	Scikit-learn	Dataset Preparation	High
3	Machine Learning Model	Random Forest / Logistic Regression	Prediction System	High

Overall Code Quality Assessment:

Aspect	Rating(1-5)	Comments
Code Layout & Structure	5	Well-organized project directory following Flask architecture.
Readability	5	Meaningful variable names, proper formatting, and comments improve readability.
Reusability	5	Modular functions and reusable ML components allow future enhancements.
Documentation / Comments	5	Adequate comments explain important sections of the code.
Overall Score	5/5	Excellent code quality with proper structure, readability, and maintainability.